FIITJEE

ALL INDIA TEST SERIES

PART TEST – II

JEE (Advanced)-2018-19

PAPER – 2

TEST DATE: 18-11-2018

Time Allotted: 3 Hours **Maximum Marks: 231**

General Instructions:

- The test consists of total 69 questions.
- Each subject (PCM) has 23 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section-C & Section-D.**

  Section-A (01 – 03, 24 – 26, 47 – 49) *contains 9 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-A (04 – 08, 27 – 31, 50 – 54) *contains 15 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 mark*** *for wrong answer.*
  Partial Marks **+1** for each correct option provided no incorrect options is selected.

  Section-A (09 – 10, 32 – 33, 55 – 56) *contains 3 paragraphs. Based upon paragraph, 2 multiple choice questions have to be answered. Each question has* ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer. There is no negative marking.*

  Section-C (11 – 20, 34 – 43, 57 – 66) *contains 30 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

  Section-D (21 – 23, 44 – 46, 67 – 69) *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Physics

PART – I

SECTION – A
Single Correct Choice Type

This section contains **3 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. A conducting disc of conductivity σ has radius a and thickness δ. A uniform magnetic field $\vec{B}$ is applied in a direction perpendicular to the plane of the disc. If the magnetic field changes with time at the rate of dB/dt, then the power dissipated in the disc due to the induced current.

(A) $\dfrac{\pi\delta\sigma a^4}{8}\left(\dfrac{dB}{dt}\right)^2$

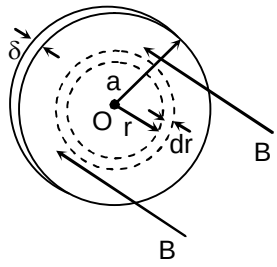
(B) $\dfrac{\pi\delta\sigma a^4}{12}\left(\dfrac{dB}{dt}\right)^2$

(C) $\dfrac{\pi\delta\sigma a^4}{4}\left(\dfrac{dB}{dt}\right)^2$

(D) $\dfrac{\pi\delta\sigma a^4}{6}\left(\dfrac{dB}{dt}\right)^2$

Ans. **A**

Sol. Consider two circles of radii r and r + dr concentric with the disc (0 < r < a) (figure). The induced e.m.f. in the circular path of radius r is



$$\varepsilon = -\frac{d}{dt}\left(\pi r^2 B\right) = -\pi r^2 \frac{dB}{dt}.$$

The resistance of the circular path between radii r and r + dr is

$$R = \frac{1}{\sigma}\frac{2\pi r}{\delta dr},$$

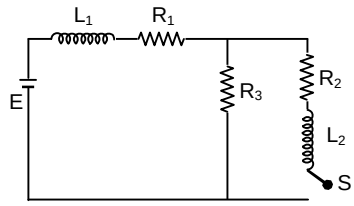
The length of the path being $2\pi r$ and the cross sectional area of current flow being δdr. The power dissipated inside this path is

$$dP = \frac{\varepsilon^2}{R} = \frac{\pi\delta\sigma}{2}\left(\frac{dB}{dt}\right)^2 r^3 dr.$$

The total dissipated power P is

$$P = \frac{\pi\delta\sigma}{2}\left(\frac{dB}{dt}\right)^2 \int_0^a r^3 dr = \frac{\pi\delta\sigma a^4}{8}\left(\frac{dB}{dt}\right)^2.$$

2. In the circuit shown in figure, the switch 'S' is closed at t = 0. The value of current in the resistor R_3, when it becomes steady (as compared to the steady current before closing the switch S)



(A) Increases

(B) Decreases

(C) Remains constant

(D) Becomes zero

Ans. B

Sol. Initial current in R_3, $i_i = \dfrac{E}{R_1 + R_3}$

Final current $i_f = \dfrac{E}{R_1 + \dfrac{R_2R_3}{R_2 + R_3}} \times \dfrac{R_2}{R_2 + R_3} = \dfrac{R_2E}{R_1(R_2 + R_3) + R_2R_3}$

$I_f = \dfrac{E}{\dfrac{R_1R_3}{R_2} + R_1 + R_3}$

As $i_f < i_i$, so current in the resistor R_3 will decrease.

3. A straight segment OP of length L of a circuit carrying current I ampere is placed along x-axis. Two infinitely long straight wires A and B each extending from z = $-\infty$ to $+\infty$ are fixed at y = $-\ell$ meter to y = $+\ell$ meter respectively. Wires A and B, each carry current I ampere along positive z-axis. Given that O is origin of the coordinate system. The magnitude of force on segment OP is

(A) $\dfrac{\mu_0}{\pi} I^2 \ell n\left(1 + \dfrac{L^2}{\ell^2}\right)$

(B) $\dfrac{\mu_0}{2\pi} I^2 \ell n\left(1 + \dfrac{L^2}{\ell^2}\right)$

(C) $\dfrac{\mu_0}{\pi} I^2 \ell n\left(1 + \dfrac{\ell^2}{L^2}\right)$

(D) $\dfrac{\mu_0}{2\pi} I^2 \ell n\left(1 + \dfrac{\ell^2}{L^2}\right)$

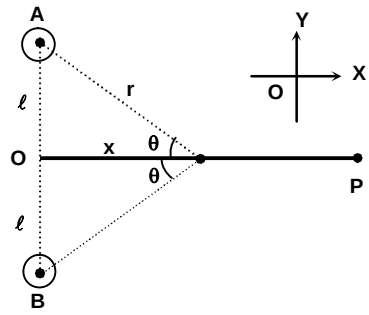
Ans. B

Sol. Net B at distance x from O along x-axis

$$= \frac{\mu_0}{2\pi}\frac{I}{r}2\cos\theta$$

$$\Rightarrow F = \int dF = \int I\frac{\mu_0}{2\pi}\frac{I}{r}2\cos\theta dx$$

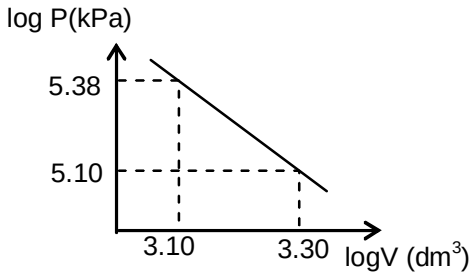
$$= \frac{\mu_0}{2\pi}I^2\ell n\left(1+\frac{L^2}{\ell^2}\right)$$



(One or More than one correct type)

This section contains **5 questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

4. Logarithms of reading of pressure and volume for an ideal gas were plotted in the graph. By measuring the gradient it can be shown that



(A) The gas may be Mono-atomic

(B) The gas may be Diatomic

(C) The process is an adiabatic change

(D) The process is an isothermal change.

Ans. **B, C**

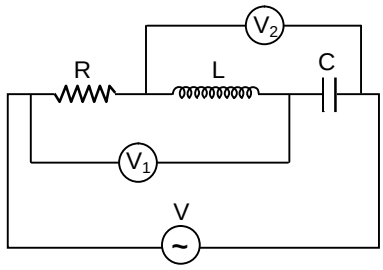
Sol. $\log P = m \log V + C$ …(i)

Where C is constant and m is slope

$$m = \left[\frac{5.38-5.10}{3.10-3.30}\right] = -1.4$$

So, from (i) $PV^{1.4} = K$ (constant)

5. In a RLC series circuit shown in the figure, the readings of voltmeters V_1 and V_2 are 100 V and 120 V respectively. The source voltage (V) of alternating source is 130 V. Mark the correct option(s).



(A) Voltage across resistor, inductor and capacitor are 50 V, 86.6 V and 186.6 V respectively.

(B) Voltage across resistor, inductor and capacitor are 50 V, 86.6, 206.6V respectively.

(C) Circuit is capacitive and power factor is $\frac{5}{13}$.

(D) Circuit is inducting and power factor is $\frac{5}{13}$.

Ans. B, C

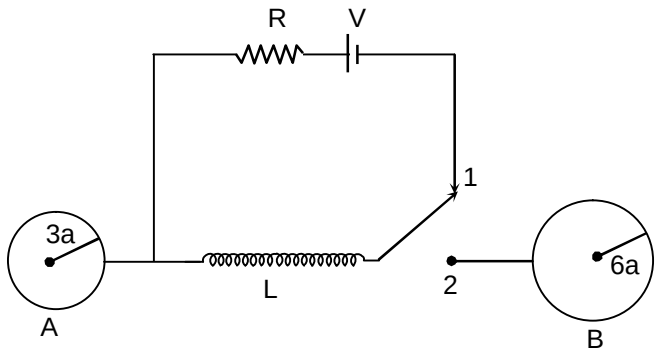
Sol. Let voltage across resistor, inductor and capacitor be V_R, V_L and V_C. Also since $V_2 > V_1$ so $V_C > V_L$.

$\therefore \quad V_R^2 + V_L^2 = V_1^2 = 100^2$ and $V_C - V_L = 120$ and $V_R^2 + (V_C - V_L)^2 = 130^2$

Solving $V_R = 50V, V_L = 86.6\,V, V_C = 206.6\,V$

Power factor = $\cos\phi = \frac{V_R}{V_Z} = \frac{50}{130} = \frac{5}{13}$

6. An electrical circuit is shown in the figure. It has two conducting spheres A and B of radii 3a and 6a respectively. When the inductor of inductance L is in its steady state the switch is shifted from position-1 to position-2. Now choose the correct option(s)



(A) Maximum charge on the sphere A is $\frac{V}{R}\sqrt{8\pi\varepsilon_0 aL}$.

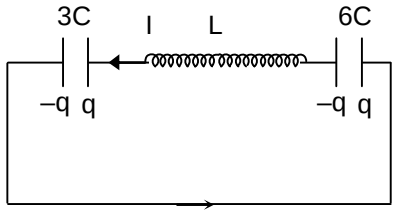
(B) Maximum charge on the sphere B is $\frac{V}{R}\sqrt{4\pi\varepsilon_0 aL}$.

(C) The circuit contains only magnetic energy at time $t = \sqrt{8\pi^3\varepsilon_0 aL}$.

(D) The circuit contains only electrical energy at time $t = \sqrt{2\pi^3\varepsilon_0 aL}$.

Ans. **A, C, D**

Sol.



$I_0 = \frac{V}{R},\ 3C = 4\pi\varepsilon_0 a \times 3 \Rightarrow C = 4\pi\varepsilon_0 a$

$\frac{dI}{dt} < 0$

$\frac{q}{3C} + \frac{q}{6C} - L\left|\frac{dI}{dt}\right| = 0$

$\Rightarrow \quad L\frac{d^2q}{dt^2} + \frac{q}{2C} = 0 \Rightarrow \frac{d^2q}{dt^2} + \frac{q}{2LC} = 0$

$\Rightarrow \quad q = q_0 \sin\omega t,\ I = q_0\omega\cos\omega t,\ \omega = \sqrt{\frac{1}{2LC}}$

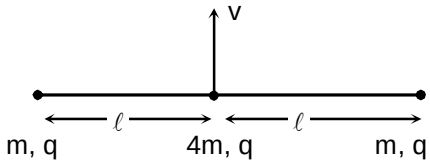
At $t = 0,\ I = I_0 \ \Rightarrow \ q_0 = \frac{I_0}{\omega} = I_0\sqrt{2LC}$

$q_0 = \frac{V}{R}\sqrt{8\pi\varepsilon_0 aL}$

$T = \frac{2\pi}{\omega} = 2\pi\sqrt{8\pi\varepsilon_0 aL}$

$\frac{T}{4} = \sqrt{2\pi^3\varepsilon_0 aL}$

7. Three small metallic charged balls (radius tends to zero) have equal charges q and masses m, 4m and m as shown in the figure. The balls are connected by light string (Non-conducting) of length ℓ each, and placed on horizontal frictionless, non-conducting table. Initially balls are at rest and form a straight line. A sharp horizontal impulse gives the central ball a speed v directed perpendicular to strings connecting the balls. Choose correct option(s):



(A) The minimum distance during subsequent motion between the balls of mass m is $\frac{6q^2\ell}{3q^2 + 16\pi m\varepsilon_0 v^2\ell}$

(B) The minimum distance during subsequent motion between the balls of mass m is $\frac{3q^2\ell}{3q^2 + 16\pi m\varepsilon_0 v^2\ell}$

(C) The maximum difference in electrostatic potential energy of the system during subsequent motion is $\frac{2mv^2}{3}$

(D) The maximum difference in electrostatic potential energy of the system during subsequent motion is $\frac{4mv^2}{3}$

Ans. **A, C**

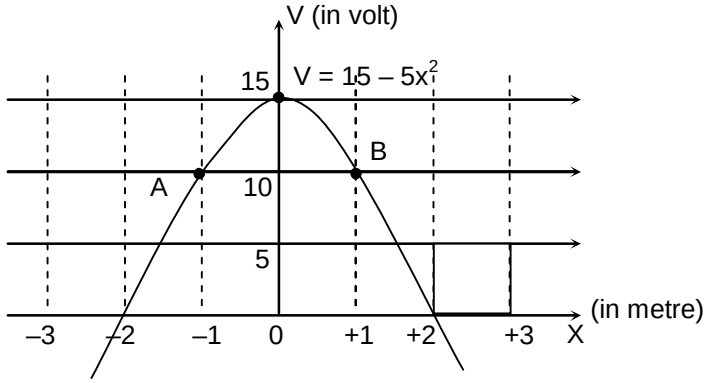
Sol. $v_{CM} = \frac{4mv}{6m} = \frac{2v}{3}$

In centre of mass frame
Gain in P.E. = Loss in KE

$$\frac{q^2}{4\pi\varepsilon_0}\left[\frac{1}{D} - \frac{1}{2\ell}\right] = 2mv^2 - \frac{4mv^2}{3}$$

$$\Rightarrow \qquad D = \frac{6q^2\ell}{3q^2 + 16m\pi\varepsilon_0 v^2\ell}$$

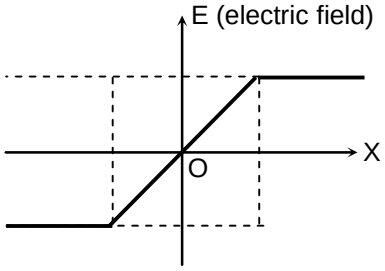
8. The figure shows the variation of electrostatic potential V in volt with the distance of position of point along x-axis from origin due to continuous volume charge distribution. In the region x = – 1m to x = + 1m, the graph is parabolic ($V = 15 - 5x^2$) and rest portion of graph is straight line. Choose the correct option(s) ($\varepsilon_0 \Rightarrow$ permittivity of free space = 8.85×10^{-12} N^{-1} $m^{-2}C^2$). The direction of $\vec{E}$ along positive x-axis is considered as positive.



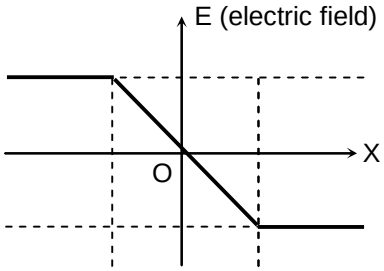
(A) This graph of potential may be due to a thick sheet of infinite dimension ($-1m \le x \le 1$, $-\infty < y < \infty$ and $-\infty < z < \infty$) with constant volume charge density 1.77×10^{-10} C/m^3.

(B) This graph of potential may be due to a thick sheet of infinite dimension ($-1m \le x \le 1$, $-\infty < y < \infty$ and $-\infty < z < \infty$) with constant volume charge density 0.885×10^{-10} C/m^3.

(C)



(D)



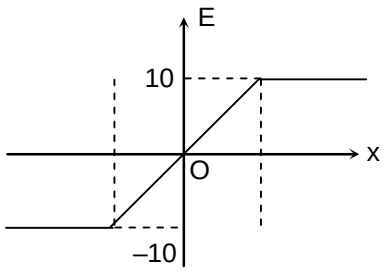
Ans. **A, C**

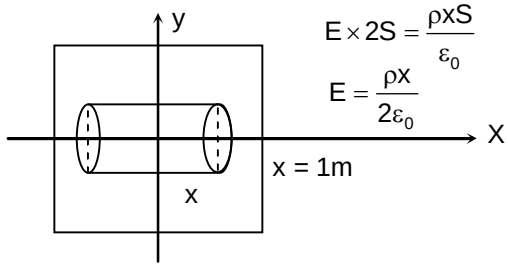
Sol. $dV = -Edx = \frac{-\rho}{2\varepsilon_0}xdx$

$\int_{V_0}^{V} dv = \frac{-\rho}{2\varepsilon_0}\int_0^x xdx \qquad \Rightarrow V = V_0 - \frac{\rho}{4\varepsilon_0}x^2$, so

$V_0 = 15$ volt

$\frac{\rho}{4\varepsilon_0} = 5 \Rightarrow \rho = 20\varepsilon_0 = 20\times 8.85\times 10^{-12} = 1.77\times 10^{-10}\ C/m^3$





$E \times 2S = \frac{\rho x S}{\varepsilon_0}$

$E = \frac{\rho x}{2\varepsilon_0}$

Paragraph type (One Option Correct)

This section contains **1 paragraph** each describing theory, experiment, data etc. **Two questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Questions 9 & 10

The melting point of wax is measured by using a thermometer which is not exact, and it indicates a temperature of 50°C as shown in figure-I.

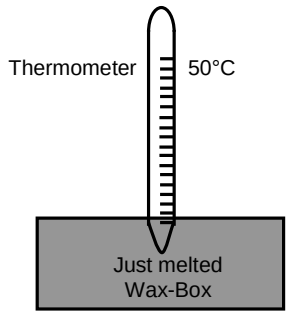


Figure-I

A Wheatstone bridge is used to measure this melting point exactly. The deviation from 50°C is being measured by observing the variation of the resistance of a coil embedded in wax.

The Wheatstone bridge is setup so that the resistance of each arm is 50Ω exactly at 50°C, and there is no current in the galvanometer as shown in figure-II. The resistance of the galvanometer is 50Ω.

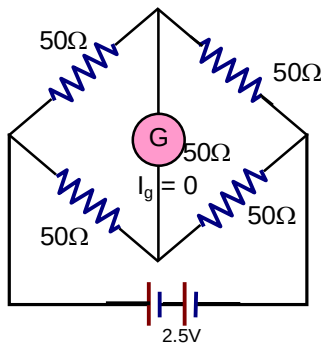


Figure-II

One arm of the bridge containing the coil of resistance 50Ω is now immersed in a wax-box. The wax is just melted as shown in the figure-III. It is now observed that the current through the galvanometer is 0.1 μA.

Given, temperature coefficient of resistance is 10^{-3} per °C. Resistances other than inside wax-box are assumed to be constant and are exactly 50Ω while the resistance of wax box is very close to 50Ω.

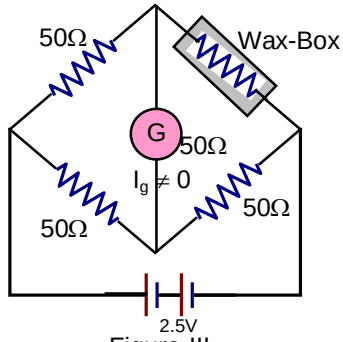


Figure-III

9. The variation in the resistance inside the wax-box from its initial value of 50Ω is

(A) $1\times10^{-4}\Omega$

(B) $2\times10^{-4}\Omega$

(C) $4\times10^{-4}\Omega$

(D) $8\times10^{-4}\Omega$

Ans. D

10. Calculate the deviation in the melting point from 50°C.

(A) 0.004°C

(B) 0.008°C

(C) 0.016°C

(D) 0.036°C

Ans. C

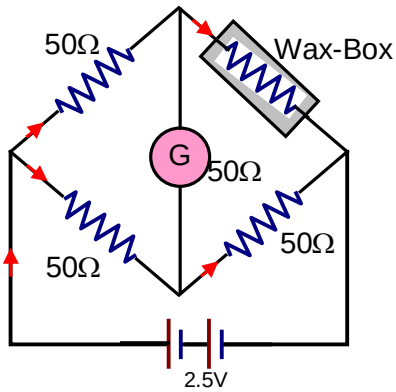
Sol. 9-10.

When the four arms have equal resistances, the off-balance galvanometer current for a small change ΔR in the resistance of the third arm is

$$I_G = \frac{V_B \Delta R}{8R^2}.$$

Here $I_G = 10^{-7}$ A, R = 50Ω, and $V_B = 2.5$V. Hence

$$\Delta R = \frac{8\times10^{-7}\times25\times10^{2}}{2.5} = 8\times10^{-4}\Omega.$$



If the corresponding change in temperature is δT, we have

$$R\alpha\Delta T = \Delta R_3,$$

Where α = temperature coefficient of resistance = 10^{-3}/°C.

Therefore, $\Delta T = \dfrac{8\times10^{-4}}{50\times10^{-3}} = 1.6\times10^{-2} = 0.016°C$.

SECTION – C
(Single digit integer type)

This section contains **10 questions**. The answer to each question is a **Single Digit Integer** ranging from 0 to 9, both inclusive.

11. An air-filled capacitor of capacitance 50 pF is charged to a voltage V and connected across another uncharged capacitor of capacitance 100 pF, the final voltage across each capacitor is V_1 volt. Now an identical air-filled capacitor of capacitance 50 pF is totally filled with a solid dielectric of dielectric constant 2n & again charged to the same voltage V as before. It is now connected across another uncharged capacitor of capacitance 100 pF, the final voltage across each capacitor is V_2 volt. If $\dfrac{V_1}{V_2}$ is $\dfrac{2}{5}$ then find the value of n.

Ans. 5

Sol. In the first case,
Charge = CV = $(C + C_e)V_1$, …(i)
Where C = 50 pF, C_e = 100 pF
In the second case,
Charge = KCV = $(KC + C_e)V_2$, …(ii)
Dividing (ii) by (i) we obtain

$$K = \frac{KC + C_e}{C + C_e}.\frac{V_2}{V_1} = \frac{KC + C_e}{C + C_e}\times 2.5 = \frac{2.5KC + 2.5C_e}{C + C_e}$$

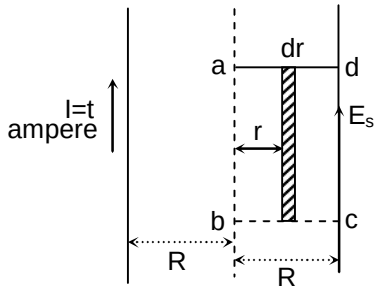
$$\Rightarrow \quad 1.5KC = (K - 2.5)C_e \quad \Rightarrow \quad 1.5K = (K - 2.5)\frac{C_e}{C} = 2(K - 2.5)$$

$$\Rightarrow \quad 0.5\,K = 5 \quad \Rightarrow \quad K = 10 \quad \Rightarrow \quad n = 5.$$

12. A long thin copper wire of the radius 2mm, carries a time-varying current I = t ampere (uniformly distributed), then the induced electric field on its surface is equal to k $X10^{-7}$ Vm^{-1}. Find the value of k. Take the induced field along the axis of the wire to be zero.

Ans. 1

Sol. Let the radius of the wire be R. Consider a rectangle abcd in the wire with the side ab along the axis. The magnetic induction at a distance r from the axis (r < R) is $B = \dfrac{\mu_0 Ir}{2\pi R^2}$



The flux through the elementary shaded area within abcd is Bldr, where ℓ = ab = cd. The flux Φ through abcd is

$$\Phi = \int_0^R \frac{\mu_0 Ir}{2\pi R^2}\ell dr = \frac{\mu_0 I\ell}{4\pi}.$$

The induced emf along the curve abcd is

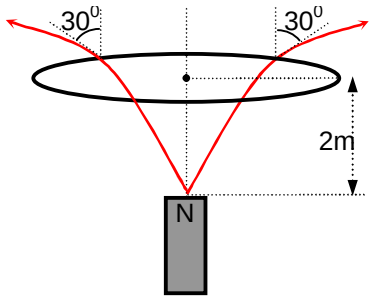
$$|\varepsilon| = \frac{d\phi}{dt} = \frac{\mu_0\ell}{4\pi}\frac{dI}{dt} = \frac{\mu_0\ell}{4\pi} \qquad \text{…(i)}$$

The wire being thin, cd >> da. If E_s is the induced electric field at the surface and E_0 that along the axis, then $|\varepsilon| = (E_s - E_0)\ell$. Since E_0 = 0, we obtain from (i)

$$E_s = \frac{\mu_0}{4\pi} = 10^{-7}\ Vm^{-1}.$$

Clearly, E_s is independent of the radius of the wire.

13. A conducting ring of radius R & mass 300 g, carrying current I ampere as shown in the figure. A bar magnet with its north pole up is placed along the symmetric axis below the conducting ring at a distance of 2m from the centre of ring. At the location of ring, the magnetic field makes an angle 30^0 with vertical and its magnitude is $\frac{1}{3\pi}$ Tesla. If ring remains at rest in its position, find the product of current I in the ring and radius R of ring in ampere-meter. $\left(g = 10\ m/s^2\right)$



Ans. **9**

Sol. $F_{mag} = mg \Rightarrow \left(B\sin 30^\circ\right) \times I \times \left(2\pi R\right) = mg$

$$\Rightarrow IR = \frac{mg}{2\pi B \sin 30^\circ} = \frac{0.3\times 10\times 3\pi \times 2}{2\pi} = 9 \text{ A-m}$$

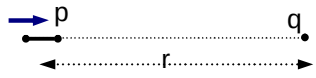
14. A neutral metal sphere is placed at a large distance from a point charge. The magnitude of Coulomb electrostatic interaction force between the charge and sphere is F_0. If the distance between them is doubled, the new magnitude of Coulomb electrostatic interaction force between the charge and sphere is $2^{-n}F_0$. Find the value of n. (Assuming that the radius of sphere is very small as compared to the distance between them)

Ans. **5**

Sol. Due to induction, the sphere behaves as dipole whose dipole moment is directly proportional to inducing field generated by point charge. So

$p = \frac{a}{r^2}$, a is a constant and r is the distance between them.

$$F_0 = \frac{bqp}{r^3} = \frac{baq}{r^5} \Rightarrow F = \frac{baq}{(2r)^5} = \frac{baq}{2^5 r^5} = 2^{-5}F_0$$

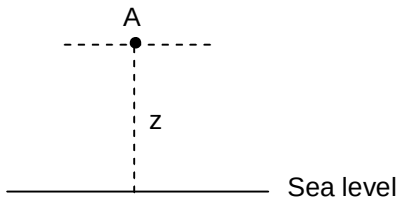


15. On a sunny day, the temperature and pressure of atmosphere varying with height z from sea level as

$$T = T_0\left(1-\frac{z}{z_0}\right), \text{ and } P = P_0\left(1-\frac{z}{z_0}\right)^{\alpha}$$

Where T_0 and z_0, are 25°C and 25 km. The temperature pressure and density of air at sea level is T_0 = 25°C, P_0 = 1 atm and 1.2 kg/m^3 respectively. Assume that air follows ideal gas law in the given condition, i.e., $\frac{\rho T}{P} = \text{constant}$.

Find the value of α.
1 atm = 10^5 N/m^2 and g = 10 m/s^2 (g is uniform)



Ans. **3**

Sol. $dp = -\rho g dz$

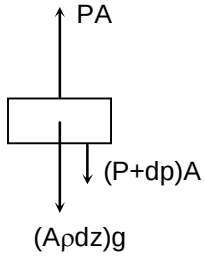
$$\Rightarrow \quad \frac{dp}{dz} = -\frac{\rho_0 g T_0 P}{P_0 T}$$

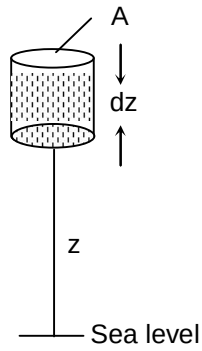
$$\Rightarrow \int_{P_0}^{P} \frac{dP}{P} = -\frac{\rho_0 g}{P_0} \int_0^z \frac{dz}{\left(1-\frac{z}{z_0}\right)}$$

$$\Rightarrow \quad \ell n\left(\frac{P}{P_0}\right) = \frac{\rho_0 g z_0}{P_0}\, \ell n\left(1-\frac{z}{z_0}\right)$$

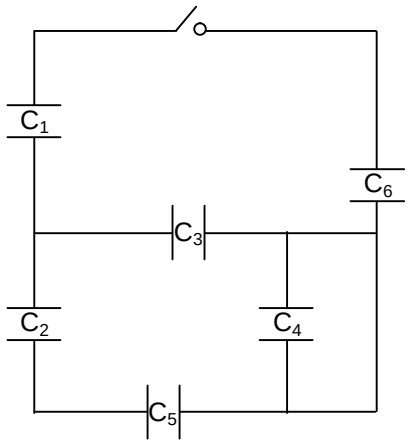
$$\Rightarrow \quad P = P_0\left(1-\frac{Z}{Z_0}\right)^{\alpha}, \text{ where}$$

$$\alpha = \frac{\rho_0 g z_0}{p_0} = \frac{1.2\times 10\times 25\times 10^3}{10^5} = 3$$



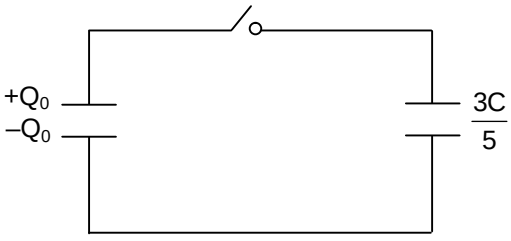


16. In the circuit shown, all capacitors are identical. Initially, the switch is open and capacitor-C_1 is the only charged capacitor. After the switch is closed, the equilibrium is re-established and the charge on the capacitor – C_1 is Q. If initial charge on the capacitor – C_1 was $Q_0 = \frac{a}{b}Q$, where a and b are positive integer, lying between 0 to 9. Find the value a – b.

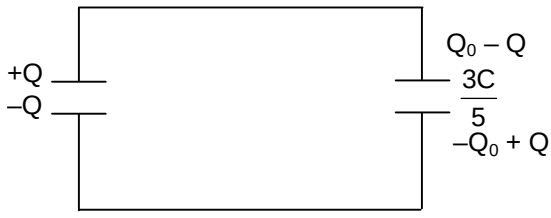


Ans. 3

Sol. $\frac{Q}{C} + \frac{-Q_0 + Q}{\frac{3C}{5}} = 0 \Rightarrow 3Q - 5Q_0 + 5Q = 0 \Rightarrow Q_0 = \frac{8Q}{5}$



Just before closing the switch



Just after closing the switch

17. A point charge Q = 500 mC and mass m = 10^{-6} kg is moving in a uniform magnetic field $\vec{B} = 1\,\text{tesla}(\hat{k})$ with speed v = 10^6 m/s and follows path as shown in figure-1. If a wire with shape same as the trajectory of charged particle carrying current I = 2A is kept in same magnetic field, each point of the wire experiences a tension T (in newton) in magnitude. Find the value of T.

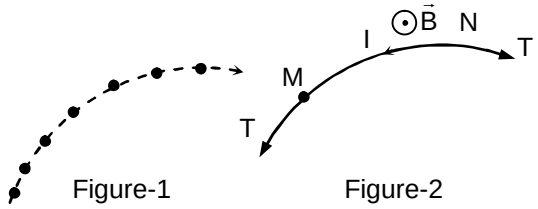


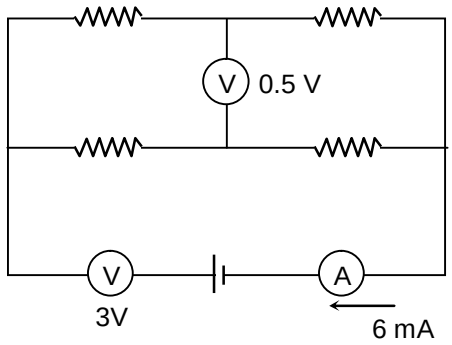
Figure-1 Figure-2

Ans. **4**

Sol. $R_1 = R_2 \Rightarrow \frac{mv}{qB} = \frac{T}{BI} \Rightarrow T = \frac{mvI}{Q}$

$\Rightarrow \quad T = \frac{10^{-6} \times 10^6 \times 2}{500 \times 10^{-3}} = 4$ newton

18. Four resistance R, 3R, R and 3R are connected as shown in figure. The two real voltmeters are identical and Ammeter is ideal. If the value of R is $25N\,\Omega$, find the value of N. You decide arrangement of the resistances so that the given conditions are satisfied.



Ans. 5

Sol. $x + y = 6 \Rightarrow x = 3.5$ mA

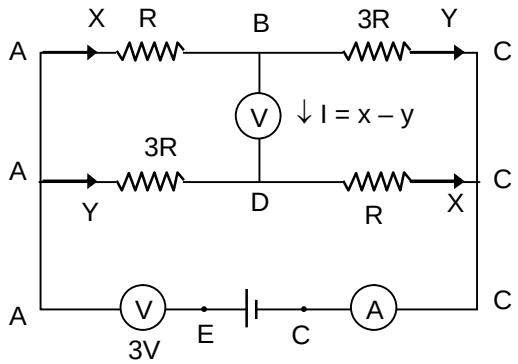
$x = 1 + y \quad y = 2.5$mA

KVL:

$-3.5R - 500 + 2.5 \times 3R = 0$

$\Rightarrow \quad 4R = 500 \Rightarrow R = 125\ \Omega$

$R_v = \frac{0.5}{10^{-3}} = 500\,\Omega$



19. Particle A with charge 2q and mass m_A, and particle B with charge 4q and mass m_B are accelerated with potential difference of V & 4 V respectively. Both particles are deflected by a uniform magnetic field into semicircular paths. The radius of trajectories of particle A and B are R and 2R respectively. The direction of magnetic field is perpendicular to the velocity. Find value of $\frac{m_B}{m_A}$.

Ans. **2**

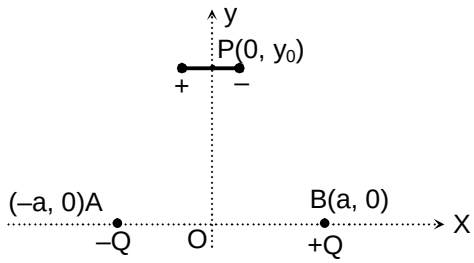
Sol. $q\Delta V = \frac{1}{2}mv^2 \Rightarrow mv = \sqrt{2qm\Delta V}$

$$R = \frac{mv}{Bq} = \frac{\sqrt{2qm\Delta V}}{Bq} \Rightarrow R \propto \sqrt{\frac{m\Delta V}{q}}$$

$$R = \sqrt{\frac{m_A\Delta V}{2q}}, \& \ 2R = \sqrt{\frac{m_B 4\Delta V}{4q}} = \frac{1}{2}$$

$$\Rightarrow \sqrt{\frac{m_A}{m_B}.\frac{1}{2}} = \frac{1}{2} \Rightarrow \frac{m_A}{m_B}.\frac{1}{2} = \frac{1}{4} \Rightarrow \frac{m_A}{m_B} = \frac{1}{2}$$

20. Two point charges +Q and –Q are kept at points (a, 0) and (–a, 0).
A dipole of dipole strength p and mass m is released from point P(0, y_0), where y_0 >>> a; as shown in the figure. If the maximum speed of dipole is $\sqrt{\frac{5Qp}{\eta\pi\varepsilon_0 ma^2}}$, find the value of η



Ans. 5

Sol. The force acting on the dipole is along negative y-axis.

$$E_i = 0, \ E_f = \frac{1}{2}mv^2 - \vec{p}.\vec{E} = \frac{1}{2}mv^2 - p\left(-\hat{i}\right).\frac{2KQ}{a^2}\left(-\hat{i}\right)$$

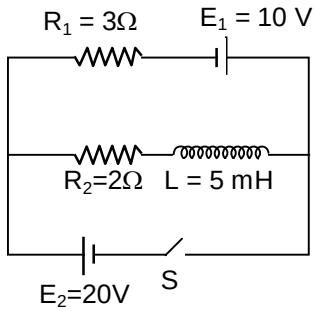
Using COME, we have

$$\frac{1}{2}mv^2 = \frac{Qp}{2\pi\varepsilon_0 a^2} \Rightarrow v = \sqrt{\frac{Qp}{\pi\varepsilon_0 ma^2}}$$

SECTION – D
(Numerical Answer Type)

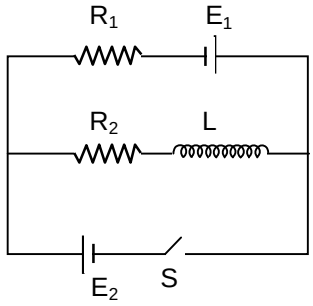
This section contains **3 questions**. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

21. Consider the circuit shown in the figure. E_1, E_2 are sources of emf and R_1, R_2 are fixed resistances. E_1 = 10V, E_2 = 20V while R_1 = 3Ω, R_2 = 2Ω. The inductance L = 5 mH.
Take $\ell n2 = 0.7$, $\ell n3 = 1.1$, $\ell n10 \approx 2.3$.
The switch S is closed for a long time. The energy stored (in joule) in the inductor is



Ans. 00000.25

Sol. After the switch S, has been closed for a long time, the current through the inductor L can be found using Kirchoff's Law: $E_2/R_2 = I_2$. The stored energy is: $\frac{1}{2}L\left(\frac{E_2}{R_2}\right)^2$.



After the switch S is opened, the current through the inductor L is given by:

$i = -I_1\left(1-e^{-R't/L}\right) + I_2 e^{-R't/L}$

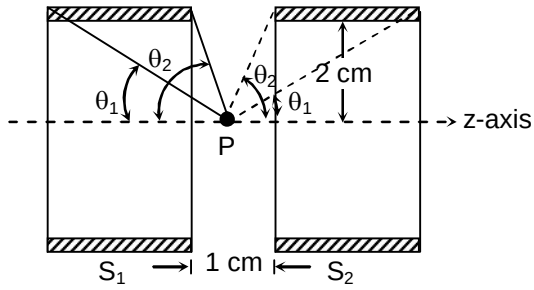
where $R' = R_1 + R_2$,

$I_1 = E_1/(R_1+R_2)$,

$I_2 = E_2/R_2$.

22. Two solenoids, each of radius 2 cm are placed coaxially with a gap of 1 cm between them. Determine the magnetic field (in Gauss) at the centre of the gap if a current of 2 A flows through the two solenoids in series. Assume that each solenoid contains 40 turns per cm.

$\cos\theta_1 = 0.907$, $\cos\theta_2 = 0.242$, $\pi = \frac{22}{7}$.



Ans. **00066.88**

Sol. Let the two solenoids be S_1 and S_2, as shown in figure. The magnetic field at the centre-point p of the gap due to the solenoid S_1 is

$$B_1 = \frac{\mu_0 NI}{2L}[\cos\theta_1 + \cos(\pi-\theta_2)] = \frac{\mu_0 NI}{2L}(\cos\theta_1 - \cos\theta_2),$$

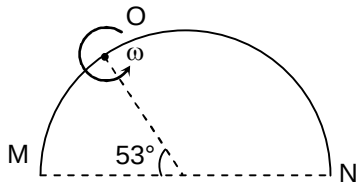
Where the angles θ_1 and θ_2 are as shown in figure. The field B_1 is along the z-axis. By symmetry, the field at p due to the solenoid S_2 has the same magnitude and direction as B_1. Thus the net magnetic field at P is

$$B = 2B_1 = \frac{\mu_0 NI}{L}(\cos\theta_1 - \cos\theta_2).$$

N/L = 40 $cm^{-1} = 4\times10^3 m^{-1}$, $\mu_0 = 4\pi\times10^{-7}$ Henry / m, and $I = 2A$.

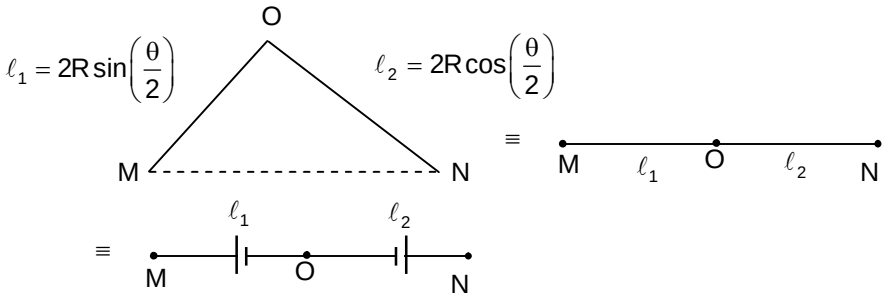
Therefore, $B = 4\pi\times10^{-7}\times4\times10^3\times2\times(0.907-.242)$ = 66.88 Gauss.

23. A semi-circular conducting wire of radius 2m is rotated in a uniform magnetic field $\vec{B} = 0.1$ Tesla $(\hat{k})$ about point O with angular speed ω = 10 rad/s as shown in the figure. The axis of rotation is parallel to $\vec{B}$. Find the potential difference (in volt) between point M and point N. $\left(\tan 53° = \frac{4}{3}\right)$.

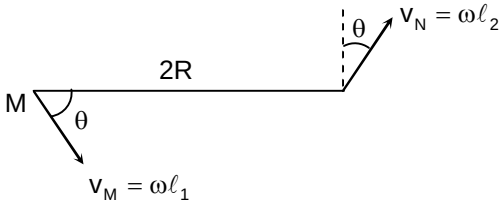


Ans. **00004.80**

Sol. $e_1 = \frac{B\omega\ell^2}{2}$, $e_2 = \frac{B\omega\ell_2^2}{2} \Rightarrow V_N - V_M = e_2 - e_1 = \frac{B\omega}{2}\left(\ell_2^2 - \ell_1^2\right) = 2B\omega R^2\cos\theta = 4.8$ volt



Second Method



$$v = \frac{1}{2}\left|\vec{V}_{N\perp} + \vec{V}_{M\perp}\right| = \omega R\cos\theta$$

$$e = B\ell v = B(2R)\omega R\cos\theta$$

$$= 2B\omega R^2\cos\theta$$

$$= 2\times 0.1\times 10\times 4\times\frac{3}{5} = 4.8 \text{ Volt}$$

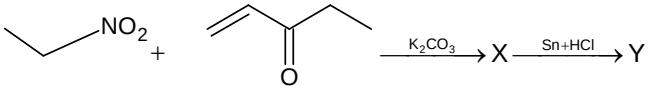
Chemistry

PART – II

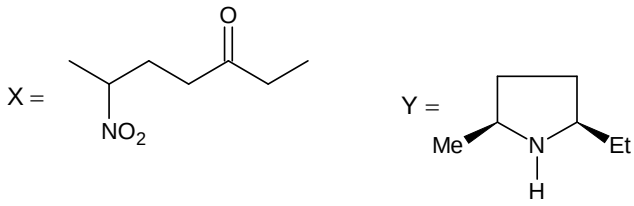
SECTION – A
Straight Objective Type

This section contains **3 multiple** choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **Only One** is correct.

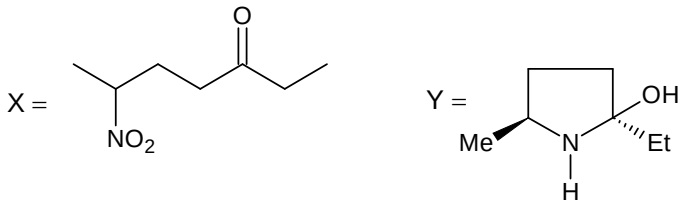
24. In the following reaction sequence the correct structure for the major products X & Y are



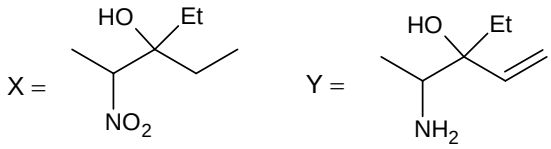
(A)



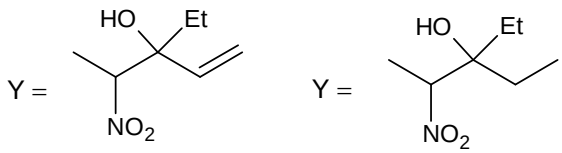
(B)



(C)

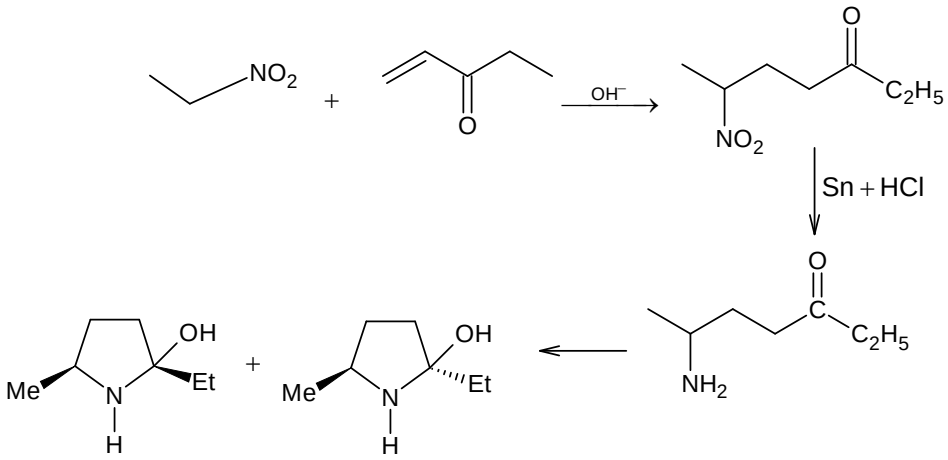


(D)



Ans. B

Sol.



25. Compound A, $C_5H_{10}O_5$ gives a tetraacetate with $\left(CH_3-\overset{O}{\overset{\|}{C}}\right)_2O$ and oxidation of A with $Br_2 - H_2O$ gives an acid, $C_5H_{10}O_6$. Reduction of A with red P and HI gives iso-pentane, there are two possible structures for compound A which can be distinguished by using

(A) HIO_4

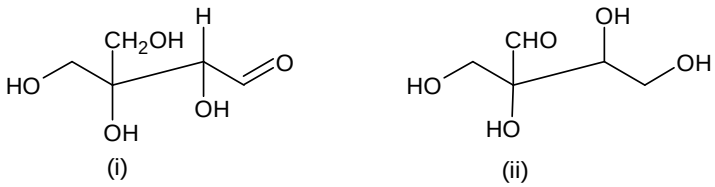
(B) Red P + HI

(C) Phenyl hydrazine

(D) Tollen's reagent

Ans. C

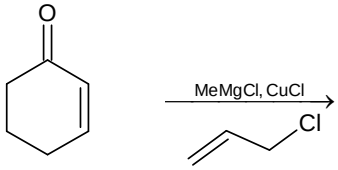
Sol.



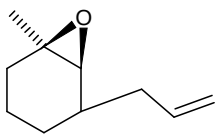
(i) (ii)

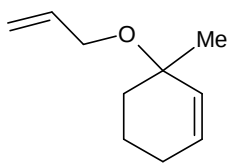
(i) will form osazone derivatives whereas (ii) will not.

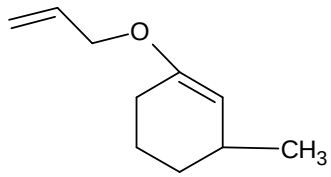
26. The major product formed in the following reaction is:

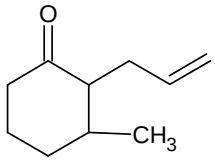


(A)

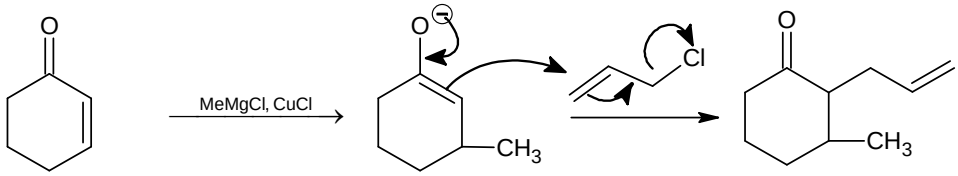


(B) 

(C) 

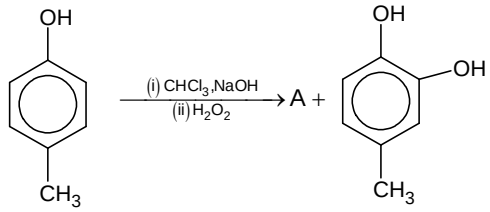
(D) 

Ans. D

Sol. 

(More Than One Correct Type)

This section contains **5 multiple** choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **MORE THAN ONE** is/are correct.

27. 

(A) A will give +ve test with Tollen's reagent

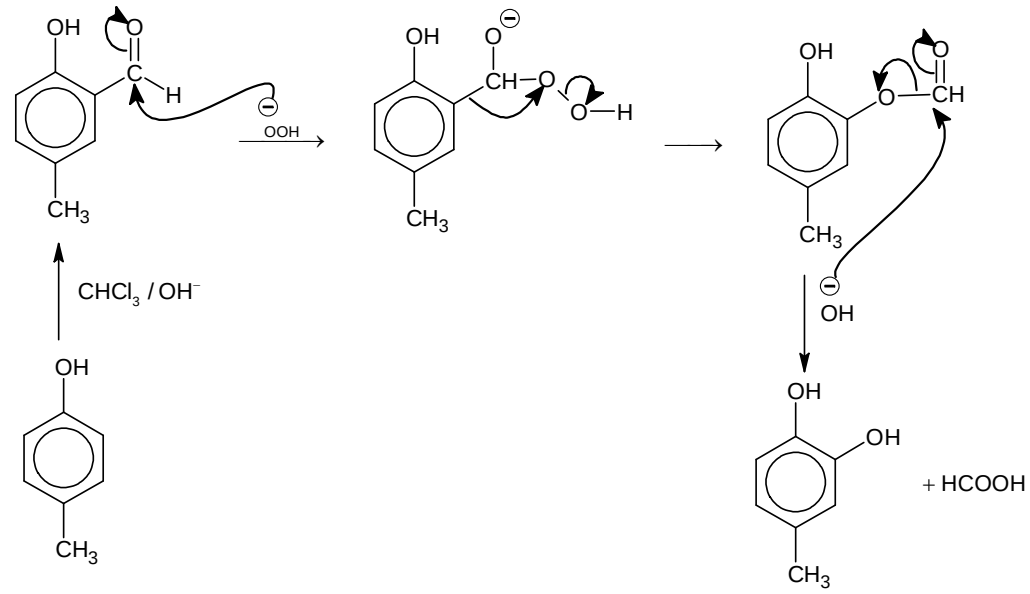
(B) A is a weak organic monobasic acid

(C) A is a aldehyde

(D) A on reaction with conc. H_2SO_4 produces a neutral gas which is a very good reducing agent.

Ans. A, B, D

Sol.



Hence A will be HCOOH.
HCOOH + conc. $H_2SO_4 \rightarrow CO$
CO is a neutral gas as well as very good reducing agent.

28. An organic compound contains C, H, S, N and Cl, for detection of chlorine, the sodium extract of compound is first heated with a few drop of conc. HNO_3, and than $AgNO_3$ is added to get a white ppt. of AgCl, why was HNO_3 added before the addition of $AgNO_3$.

(A) To prevent the formation of NO_2.

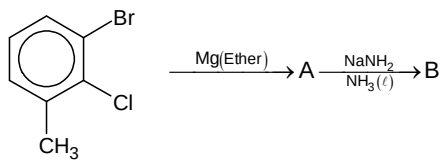
(B) To create a common ion effect.

(C) To convert CN^- and S^{-2} to volatile HCN and H_2S, else they will interfere with the test forming AgCN and Ag_2S.

(D) To prevent the hydrolysis to NaCN and Na_2S.

Ans. C

29.



(A) Both A and B are aromatic in nature.

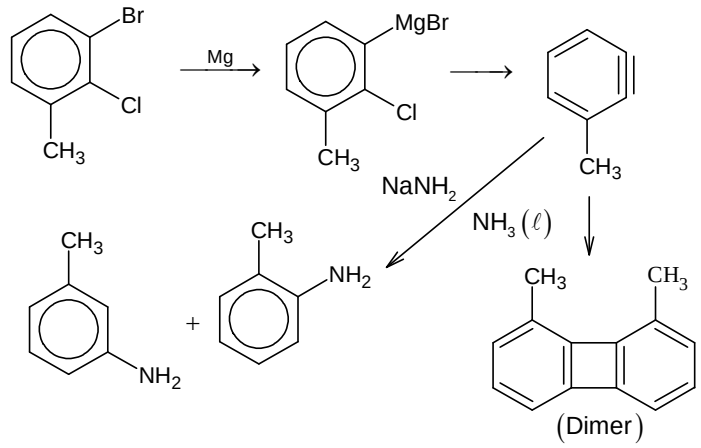
(B) Product A can show nucleophilic addition reaction.

(C) If potential nucleophile is not present then A will form dimer.

(D) B is a mixture of two structural isomers.

Ans. A, B, C, D

Sol.



30. $CH_3 - CH_2 - NO_2(X) \xrightarrow{Sn+HCl} K(\text{Organic compound})$, then which of the following is/are true for K?

(A) K (form solid oxamide) with diethyl oxalate.

(B) K (form liquid oxamide) with diethyl oxalate.

(C) K will not give Liberman nitroso test.

(D) K will form blue colour complex with Cu^{+2} salt.

Ans. A, C, D

Sol. $CH_3 - CH_2 - NO_2 \xrightarrow{Sn+HCl} CH_3 - CH_2 - NH_2 [K]$

1° amine form solid oxamide

2° amine form liquid oxamide

Liberman Nitroso test is given by 2° amine

$Cu^{+2} + 4C_2H_5NH_2 \longrightarrow \underset{(\text{Blue})}{\left[Cu(C_2H_5NH_2)_4\right]^{+2}}$

31. In which case charge in transition state is more dispersed than starting material?

(A) $CH_3X + NaNH_2$

(B) $CH_3X + NH_3$

(C) $(CH_3)_4N^+X^- + OH^-$

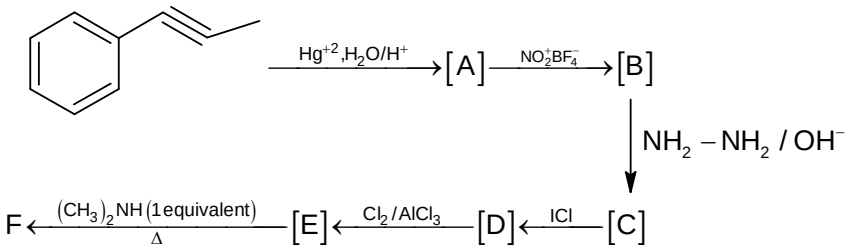
(D) $(CH_3)_4N^+X^- + RSH$

Ans. A, C, D

Paragraph type (One Option Correct)

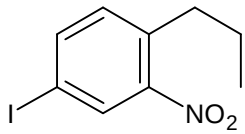
This section contains **1 paragraph** each describing theory, experiment, data etc. **Two questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Question Nos. 32 to 33

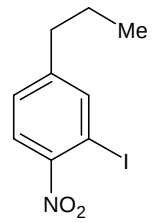


32. The compound (D) is

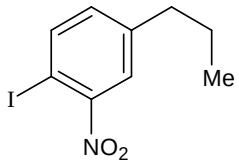
(A)



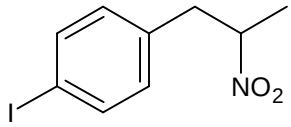
(B)



(C)

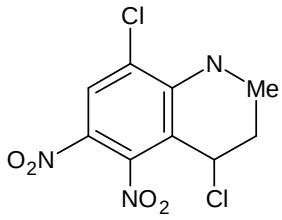


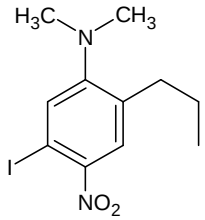
(D)

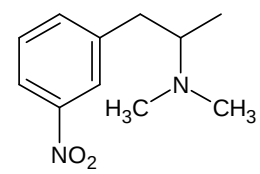


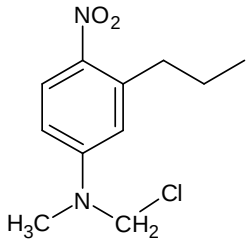
Ans. C

33. The compound F is

(A) 

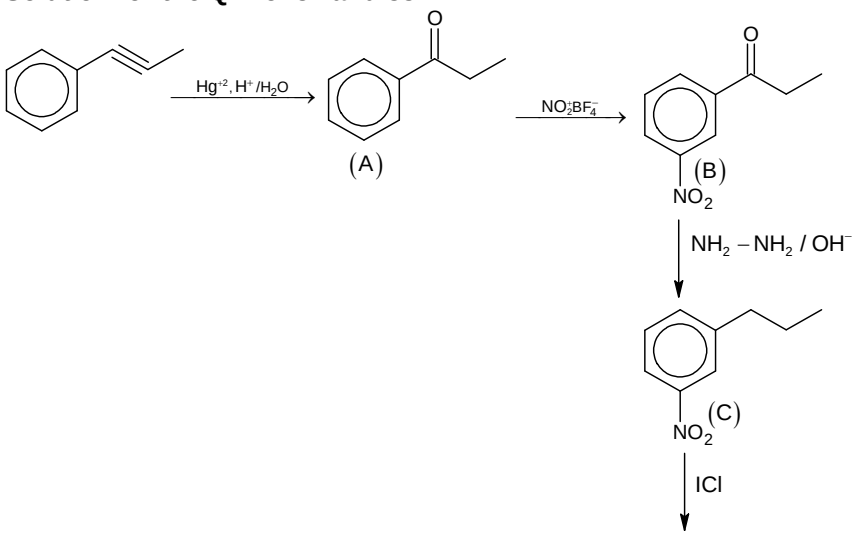
(B) 

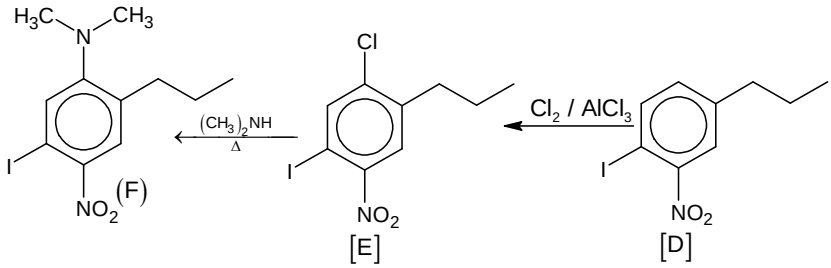
(C) 

(D) 

Ans. B

Solution for the Q. No. 32 and 33.





Single Integer Answer Type

This section contains **10 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

34. 2.79 gm of nitrogenous compound on reaction with CS_2 and $HgCl_2$ forms a black ppt. of 0.09 mol. If same amount of nitrogenous compound is reacted with HNO_2. How many litres of gas at STP would be formed.

Ans. 2

Sol. $\underset{0.09}{RNH_2} + CS_2 + HgCl_2 \longrightarrow RNCS + \underset{0.09}{HgS}\downarrow$

$\underset{0.09}{RNH_2} + HNO_2 \longrightarrow ROH + \underset{0.09}{N_2}\uparrow$

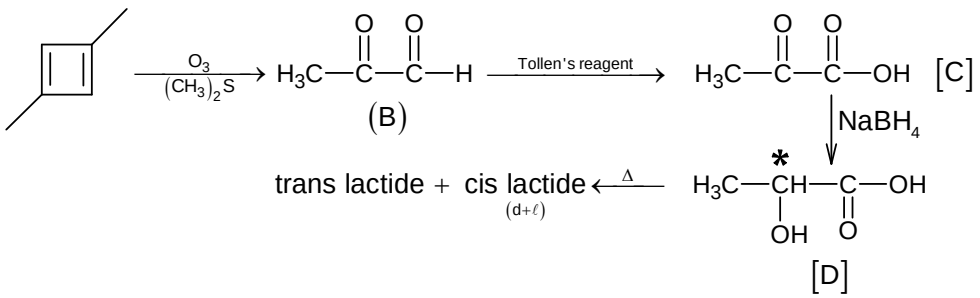
Volume in litres at STP = $0.09 \times 22.4 = 2.016 \cong 2$

35. $C_6H_8\,(\text{anti aromatic}) \xrightarrow{O_3,(CH_3)_2S} \underset{\text{Only product}}{B} \xrightarrow{\text{Tollen's reagent}} C \xrightarrow{NaBH_4} \underset{\text{Optically active}}{D}$

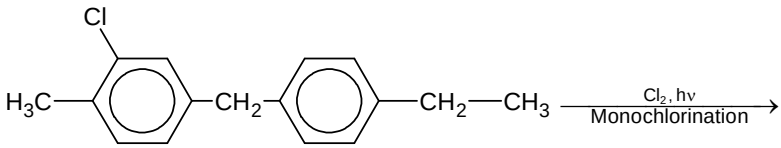
D on heating gives a major product having x number of stereoisomer, x is:

Ans. 3

Sol.



36.



If x = maximum number of monochloroisomer possible

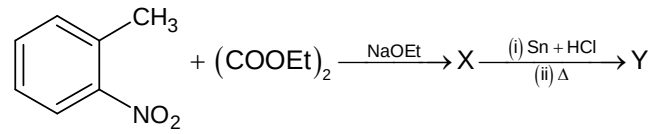
y = Monochloro isomer on fractional distillation will separate. Then find $\frac{(x+y)}{2}$

Ans. 5

Sol. x = 6

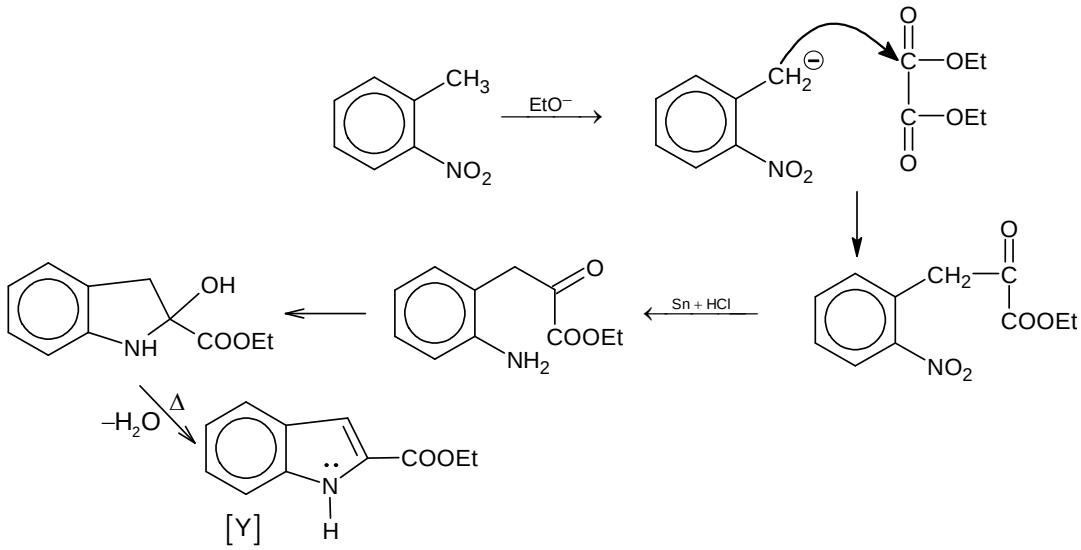
$y = 4$

$$\frac{(x+y)}{2} = 5$$

37. 

Find the number of π-bond in the Y?

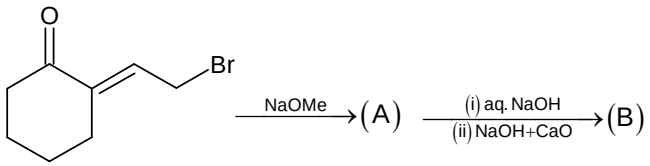
Ans. 5

Sol. 

38. How many of the following are cross-linked polymer?
Bakelite, Urea-formaldehyde, Melamine-formaldehyde, Nylon-6,6, Terylene.

Ans. 3

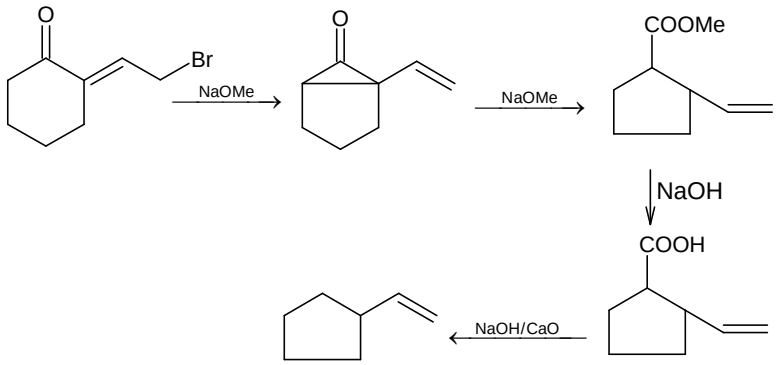
Sol. Bakelite, Urea-formaldehyde, Melamine-formaldehyde.

39. 

The number of π-bond in product (B) is:

Ans. 1

Sol.



40. 1.2 gram of acetophenone on reaction with 76.2 gram of I_2 in presence of NaOH give solid iodoform with 75% yield. Find the weight of iodoform formed in gram.

Ans. 3

Sol. $Ph-\overset{\overset{O}{\|}}{C}-CH_3 + 3I_2 + 4NaOH \longrightarrow CHI_3 + Ph-\overset{\overset{O}{\|}}{C}-ONa$

1 mole of $Ph-\overset{\overset{O}{\|}}{C}-CH_3$ will give 1 mole of CHI_3

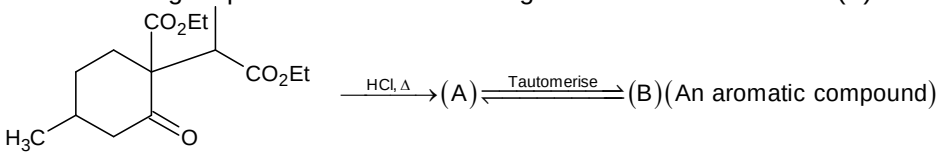
120 gram will give $\equiv$ 394 g CHI_3

1.2 gram will give $\equiv$ 3.94 g CHI_3

Hence, with 75% yield $= 3.94 \times \frac{75}{100} = 2.955$ gram

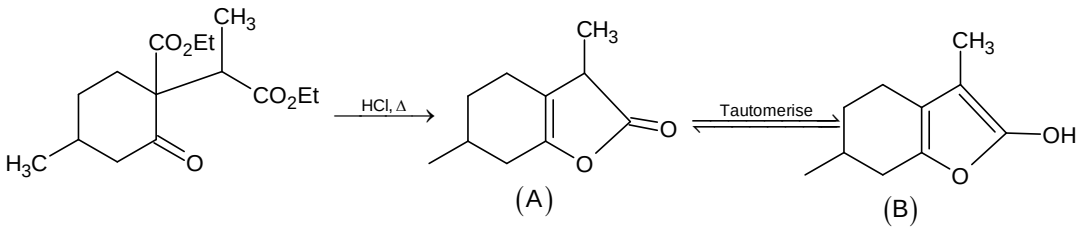
$\cong$ 3 gram.

41. In the following sequence of reaction the degree of unsaturation in the (B) is



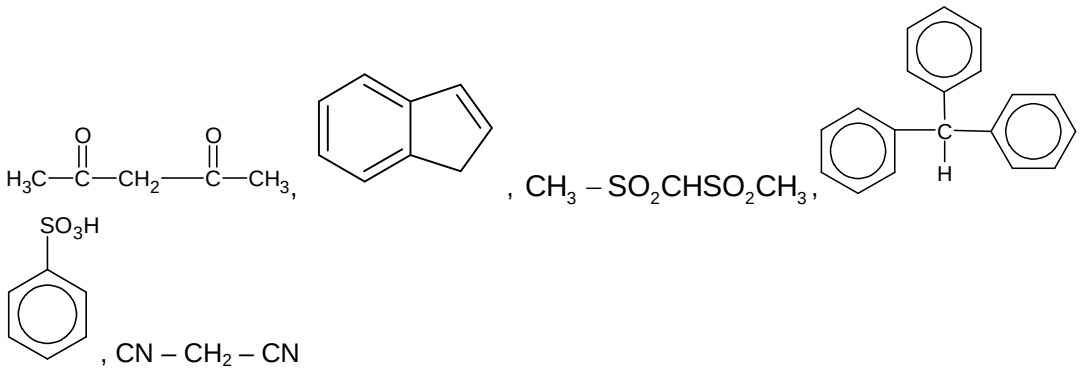
Ans. 4

Sol.

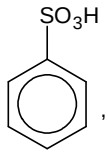


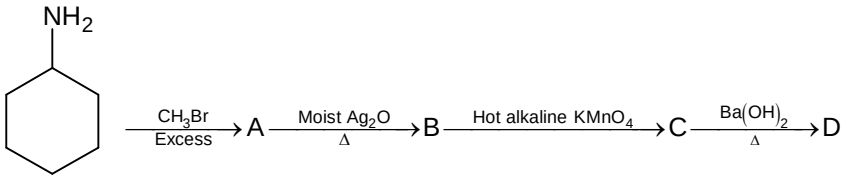
Degree of unsaturation = 4

42. How many will liberates CO_2 gas on reaction with $NaHCO_3$?



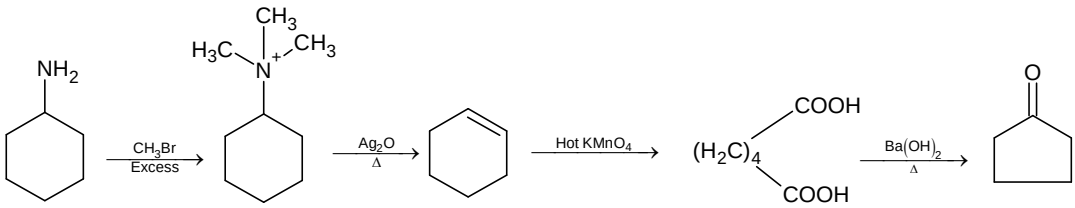
Ans. 1

Sol.  , Rest all have pK_a value greater than H_2CO_3.

43. 

If number of carbon atom in D is x and π-bond is y.
Find x + y?

Ans. 6

Sol. 

SECTION – D
(Numerical Answer Type)

This section contains **3 questions**. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

44. 2.25 g of a nitrogenous compound having molecular formula (C_2H_7N) is reacted with 11.95 g of $CHCl_3$ in presence of alc. KOH to form a bad smelling gas. One of the remaining reactant after some times becomes poisonous due to aerial oxidation. This poisonous gas is now removed by C_2H_5OH to form a precipitate. Find the weight of precipitate (in gram).

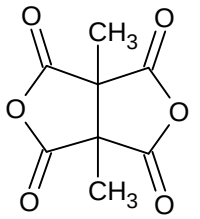
Ans. 00005.90

Sol. $\underset{\frac{1}{20}}{C_2H_5NH_2} + \underset{\frac{1}{10}}{\underset{\frac{1}{20}}{CHCl_3}} \xrightarrow{\text{alc KOH}} \underset{\frac{1}{20}}{C_2H_5NC}$

$CHCl_3 \xrightarrow{O_2,\, h\nu} COCl_2$

$COCl_2 + 2C_2H_5OH \longrightarrow (C_2H_5)_2CO_3$

Hence precipitate = $\frac{1}{20} \times 118 = 5.9$ gram

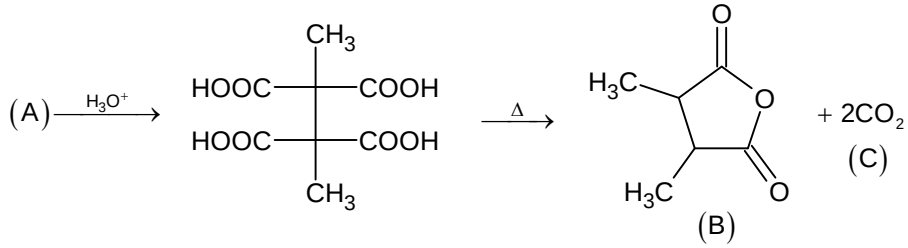
45. 

$\xrightarrow[(ii)\,\Delta]{(i)\,H_3O^+} (B) + (C)$

(A)

1.1 mole of (A) was reacted according to above reaction. 'C' is the gas produced which turns lime water milky. The amount of C formed in gram will be? (Given At. Wt. of C, O, H are 12, 16, 1 gm respectively).

Ans. 00096.80

Sol. 

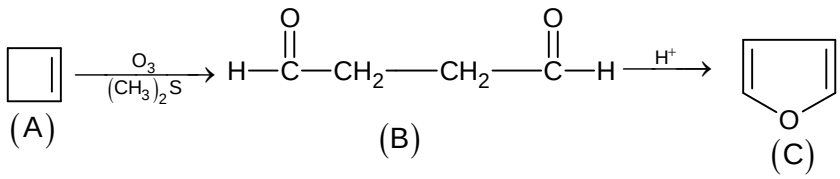
1.1 mole of (A) produce 2.2 mole of CO_2(g)

$\therefore$ wt. of $CO_2 = 2.2 \times 44 = 96.8$ gm.

46. $\underset{\text{6.75 gram}}{A(C_4H_6)} \xrightarrow[(CH_3)_2S]{O_3} B \xrightarrow[(ii)\,\Delta]{(i)\,H^+} \underset{\text{Aromatic compound}}{C}$

A is a cyclic isomer having molecular formula C_4H_6. How many gram of C would be formed if 6.75 g of A was taken?

Ans. 00008.50

Sol. 

(A) (B) (C)

Moles of A $= \frac{6.75}{54} = \frac{1}{8}$ mole

Hence, $\frac{1}{8}$ mole of C would be formed

Wt. of C $= \frac{1}{8} \times 68 = 8.5$ gm

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **3 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D), out of which only **one** is correct

47. The length of medians of a triangle are 3, 4 and 5 cm. The area (in square cm) of the triangle is

(A) 8

(B) 12

(C) 24

(D) 32

Ans. **A**

Sol. $A = \frac{4}{3} \times \frac{1}{2} \times 3 \times 4 = 8$

48. The number of right-angled ΔABC ($\angle A = 90^\circ$) with integers side length whose inradius is 3 are

(A) 3

(B) 4

(C) 5

(D) 8

Ans. **A**

Sol. $\because \ (s-a)\tan\left(\frac{A}{2}\right) = r \ \ (\angle A = 90^\circ)$

$(s - a) = r$, $2s - 2a = 2r$, $b + c - a = 2r$, $a = (b + c - 2r)$

$(b - 6)(c - 6) = 2 \times 3^2$

49. P is a point inside an equilateral triangle ABC such that PA = 9, PB = 12 and PC = 15, then side length of the ΔABC is

(A) $\sqrt{225 + 108\sqrt{5}}$

(B) $\sqrt{225 - \sqrt{33}}$

(C) $\sqrt{49 + 5\sqrt{2}}$

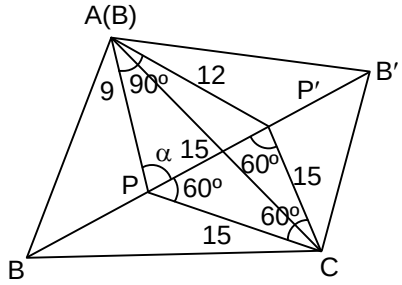
(D) none of these

Ans. A

Sol. $\cos(\alpha+60^o)=\dfrac{15^2+9^2-(AC)^2}{2.15.9}$ ..... (1)

Also, $\cos\alpha=\dfrac{9}{15}=\dfrac{3}{5}$

So, $a=\sqrt{225+108\sqrt{5}}$



One OR More Than One Choice Type

This section contains **5 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **one or more than one** is/are correct

50. Which of the following are **INCORRECT?**

(A) $(\sin x)^{\sin x} > (\cos x)^{\cos x}\ \forall\ x \in \left(0, \dfrac{\pi}{4}\right)$

(B) $(\cos x)^{\sin x} < (\sin x)^{\cos x}\ \forall\ x \in \left(0, \dfrac{\pi}{12}\right)$

(C) $x\cot x < \dfrac{\sqrt{3}}{\pi}\ \forall\ x \in \left(0, \dfrac{\pi}{3}\right)$

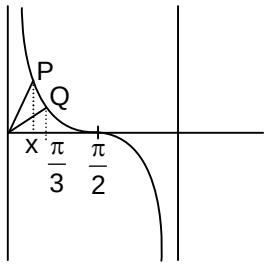
(D) $\sqrt{\sin x+\cos 2x} < \sqrt{\sin x}+\sqrt{\cos 2x}\ \forall\ x \in \left(0, \dfrac{\pi}{4}\right)$

Ans. A, B, C

Sol. (A) Use Jensen's inequality for $\sin x < \cos x < \sin x + \cos x$ with weights
$\lambda_1 = \tan x$ and $\lambda_2 = 1 - \tan x$ ($\because\ 0 < x < \pi/4$), λ_1 and λ_2 are positive)
So, $\log(\cos x) = \log(\tan x \sin x + (1 - \tan x)(\sin x + \cos x))$
$> \tan x \log(\sin x) + (1 - \tan x)\log(\sin x + \cos x)$
$\because$ $\sin x + \cos x > 1$ and $\tan x < 1$. So, $\log(\cos x) > \tan x \log(\sin x)$

(B) For $x \in \left(0, \dfrac{\pi}{12}\right)$; $\cos x > \sin x$. So, $\ln\dfrac{(\cos x)}{\cos x} > \ln\dfrac{(\sin x)}{\sin x}$

(C) Shown in the figure
m(OP) > m(OQ)



(D) $\because$ $(a + b)^p \le a^b + b^p$ for $a, b > 0$ and $0 \le p \le 1$

51. Which of the following are **INCORRECT?**

(A) $\frac{1}{3} > \sin 20^o > \frac{7}{20}$

(B) $\cos x < 1 - \frac{x^2}{2} + \frac{x^4}{16} \ \forall\, x \in \left(0, \frac{\pi}{2}\right)$

(C) $\tan x < x + \frac{x^3}{3} + \frac{2x^5}{15} \ \forall\, x \in \left(0, \frac{\pi}{2}\right)$

(D) cos 2º and cos 7º are irrational

Ans. A, C

Sol. Use $\sin 3t = 3 \sin t - 4 \sin^3 t$
Let $t = 20^o$

52. Let $x - 2y - 5 = 0$ be the directrix of a parabola and $x - y - 7 = 0$ be the tangent drawn to the parabola at the point P(4, –3), then which of the following are **INCORRECT?**

(A) length of latus rectum = $\frac{16}{\sqrt{5}}$

(B) harmonic mean of length of segments of the focal chord = $\frac{8}{\sqrt{5}}$

(C) intersection point of the directrix and the given tangent is (9, 2)

(D) circle drawn on assuming P(4, –3) and Q as diameter, always passes through focus of the parabola (where Q is the intersection point of tangent and directrix)

Ans. A, B

Sol. Foot of perpendicular P(4, –3) to the directrix is (3, –1)

Now, find its image about the tangent $\frac{\alpha - 3}{1} = \frac{\beta + 1}{-1} = \frac{-2(3 + 1 - 7)}{2}$

$\alpha = 6$, $\beta = -4$. So, focus $\equiv (6, -4)$

Now, distance of focus from the directrix = $\frac{|6 + 8 - 5|}{\sqrt{5}} = \frac{9}{\sqrt{5}}$

53. Let ABCD be a quadrilateral inscribed in a circle. Suppose $AB = \sqrt{2 + \sqrt{2}}$ and AB subtends 135º at the centre of the circle, the maximum possible area of ABCD is

(A) $\frac{5 + 3\sqrt{3}}{4\sqrt{2}}$

(B) $\frac{8 + \sqrt{7}}{\sqrt{3}}$

(C) $\frac{19+\sqrt{3}}{2}$

(D) none of these

Ans. A

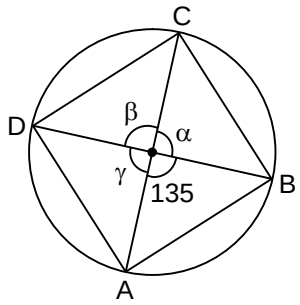
Sol. $\because$ $\alpha, \beta, \gamma < \pi$ and $y = \sin x$ is concave downward on $(0, \pi)$

So, $\frac{\sin\alpha+\sin\beta+\sin\gamma}{3} \le \sin\left(\frac{\alpha+\beta+\gamma}{3}\right) = \sin 75^o$

So, area of ABCD $\le \frac{1}{2}\sin(135^o) + \frac{1}{2}(\sin\alpha+\sin\beta+\sin\gamma)$

$= \frac{5+3\sqrt{3}}{4\sqrt{2}}$

Equality holds if $\alpha = \beta = \gamma = 75^o$



54. If the vertices of an equilateral Δ lies on the curve $8x^3 + y^3 + 6xy = 1$, then

(A) side length of the triangle is $\frac{6}{\sqrt{15}}$

(B) side length of the triangle is $\frac{\sqrt{3}}{2}$

(C) area of the triangle is $\frac{3\sqrt{3}}{5}$

(D) area of the triangle is $\frac{3\sqrt{3}}{16}$

Ans. A, C

Sol. $(2x + y - 1)(4x^2 + y^2 + 1 - 2xy + y + 2x) = 0$ and side length of the $\Delta = \frac{6}{\sqrt{15}}$

So, area = $\frac{3\sqrt{3}}{5}$

Paragraph type (One Option Correct)

This section contains **1 paragraph** each describing theory, experiment, data etc. **Two questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Question Nos. 55 to 56

Read the following write up carefully and answer the following questions:

Diameter of ellipse is a line passing through the centre of the ellipse. Consider the set of parabola(s) having common chord of minimum length with the diameter of the ellipse $x^2 + y^2 + xy = 8$ and the line $mx - y + 4 = 0$ as directrix

55. The set of value(s) of m for which only one parabola is possible is

(A) $\left\{-\sqrt{2}, \sqrt{2}\right\}$

(B) $\left\{-\sqrt{5}, \sqrt{3}\right\}$

(C) {–2, 2}

(D) none of these

Ans. A

56. The range of value(s) of m for which two parabola(s) are possible is

(A) $\left(-\infty, -\sqrt{2}\right) \cup \left(\sqrt{2}, \infty\right)$

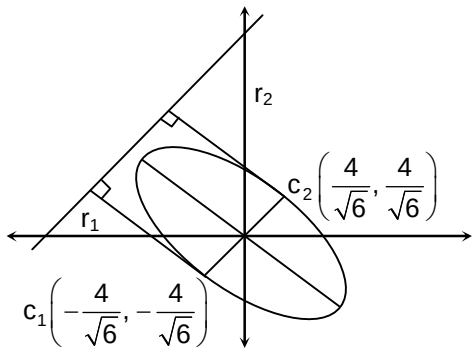
(B) $(-\infty, -2) \cup (2, \infty)$

(C) $\left(-\infty, -\sqrt{3}\right) \cup \left(\sqrt{3}, \infty\right)$

(D) none of these

Ans. A

Sol. (55.-56.)
For the ellipse $x^2 + y^2 + xy = 8$, $c \equiv (0, 0)$
For one parabola $\frac{8}{\sqrt{m^2+1}} = \frac{8}{\sqrt{3}}$: ($\because$ $C_1C_2 = r_1 + r_2$)
For two parabola(s) : $C_1C_2 < r_1 + r_2$



SECTION – C

(One Integer Value Correct Type)

This section contains **10 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

57. Consider the ellipse $4x^2 + y^2 - 8x + 4y + 4 = 0$. Pair of tangents PT_1 and PT_2 are drawn to the ellipse from P(–8, –2). Let S_1' be the image of S_1 about PT_1 and S_2' be the image of S_2 about PT_2, then if $\angle PS_1S_2' = \alpha$ and $\angle PS_1'S_2 = \beta$, then $\frac{\alpha}{\beta}$ is equal to _____

Ans. 1

Sol. $\Delta PS_1'S_2$ and $\Delta PS_1S_2'$ are similar

58. If tangents drawn to the parabola at A(1, 2) and B(5, 8) intersect at (–1, 8), then slope of directrix is $\frac{a}{b}$ where gcd(a, b) = 1, then a + b is _____

Ans. 7

Sol. Q = mid-point of AB = (3, 5), let P ≡ (–1, 8), m(PQ) = $-\frac{3}{4}$

Slope of directrix = $\frac{4}{3}$

59. Given a point (3, 2), then the minimum perimeter of a triangle with one vertex at (3, 2) one on the x-axis and one on the line y = x (you may assume that a triangle with minimum perimeter exists) is $\sqrt{\alpha}$, then sum of digits of α is _____

Ans. 8

Sol. For minimum perimeters (2, 3), (d, d) (c, 0), (3, –2) are collinear
Hence, mini perimeter = $\sqrt{26}$

60. If focus of the parabola which touches x = 0, y = 0, x – y + 1 = 0 and –2x – y – 8 = 0 is (α, β), then $\left|\frac{\alpha}{\beta}\right|$ is _____

Ans. 3

Sol. Focus is intersection point of circles x(x + 1) + y(y – 1) = 0 and (x)(x + 4) + (y)(y + 8) = 0
≡ $\left(-\frac{6}{5}, \frac{2}{5}\right)$

61. If $3\sqrt{2}$, $4\sqrt{2}$ and $\frac{12}{5}\sqrt{2}$ be the altitudes of a triangle, then $r\sqrt{2}$ (r is inradius of the triangle) is equal to _____

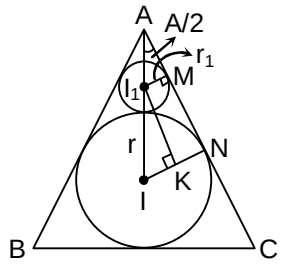
Ans. 2

Sol. ∵ Harmonic mean of the altitudes of a triangle is 3 times its inradius
So, $r = \frac{1}{19}\left(\frac{5}{12\sqrt{2}} + \frac{1}{4\sqrt{2}} + \frac{1}{3\sqrt{2}}\right)$

62. Let ABC be a triangle (with $\angle A = \pi/3$) and a circle C_1 be drawn lying in side the triangle, touching its in-circle externally and also touching the two sides AB and AC, then ratio of the radii of the circles C_1 and C is k, then 3k is equal to _____

Ans. 1

Sol. In triangle II_1K



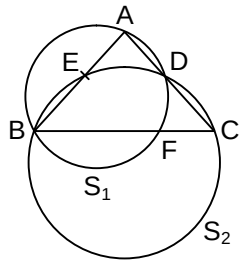
$$\sin\left(\frac{A}{2}\right)=\frac{IK}{II_1}=\frac{r-r_1}{r+r_1}\Rightarrow\frac{1-\sin\left(\frac{A}{2}\right)}{1+\sin\left(\frac{A}{2}\right)}=\frac{r+r_1-r+r_1}{r+r_1+r-r_1}$$

$$\tan^2\left(\frac{\pi-A}{4}\right)=\frac{r_1}{r}.\ \text{So, }\frac{r_1}{r}=\tan^2(30^o)=\frac{1}{3}$$

63. Let BD be the internal angle bisectors of angle B in triangle ABC with D on side AC. The circumcircle of triangle BDC meets AB at E, while the circumcircle of triangle ABD meets BC at F. if AE = 3, then CF is equal to

Ans. 3

Sol. Let circumcircle of triangle BDC be S_1 and the circumcircle of triangle ABD be S_2



So, $\frac{AD}{CD}=\frac{AB}{BC}$ ..... (1)

Now, for circle S_1:

$AE\times AB = AD\times AC$

$\Rightarrow AE=\frac{AD\times AC}{AB}$ ..... (2)

For circle S_2:

$CF\times CB = CD\times CA$

$\Rightarrow CF=\frac{CD\times CA}{CB}$ ..... (3)

From equation (2) and (3), we get

$\frac{AE}{CF}=\frac{AD\times CB}{AB\times CD}$ ..... (4)

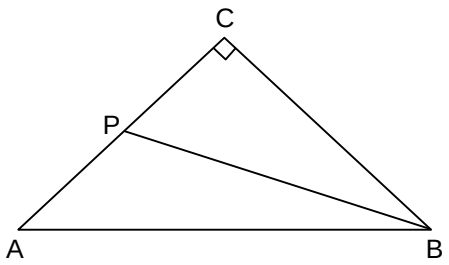
From equation (1) and (4), AE = CF

So, CF = 3

64. In an isosceles right angled triangle ΔABC, CA = CB = 1, and P is an arbitrary point on the perimeter of ΔABC, then if maximum value of PA·PB·PC is α, then $6\sqrt{2}\alpha$ is _____

Ans. 3

Sol. **(I):** Let $P\in AC$



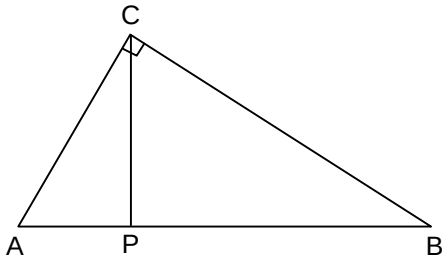
$PA\cdot PC\le\frac{1}{4}$ and $PB\le\sqrt{2}$

Thus $PA\cdot PC\cdot PB\le\frac{\sqrt{2}}{4}$

The two equality sign can not be valid at the same time

So, $PA\cdot PC\cdot PB<\frac{\sqrt{2}}{4}$

(II): $P \in AB$



$AP = x \in \left(0, \sqrt{2}\right]$

So, let $f(x) = (PA)^2(PB)^2)PC)^2$

$= x^2\left(\sqrt{2}-x\right)^2\left(1+x^2-x\sqrt{2}\right)$

Let $t = x\left(\sqrt{2}-x\right) : t \in \left[0, \frac{1}{2}\right]$

So, $f(x) = g(t) = t^2(1-t)$

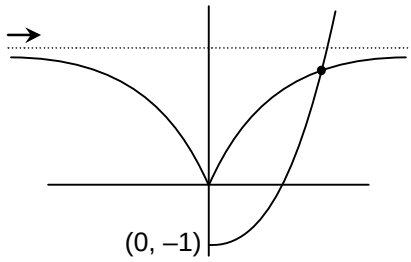
$\therefore\ f(x) \le g(t) \le g\left(\frac{1}{2}\right) = \frac{1}{8}$

$PA \cdot PB \cdot PC \le \frac{1}{2\sqrt{2}}$. Clearly P is the mid-point of AB

65. Number of solution of the equation $\left(\cos^{-1}\left(\frac{1-4x^2}{1+4x^2}\right)\right)^{\frac{3}{4}} = x^2 - x\sin x - \cos x$ is

Ans. 2

Sol. Shown in the figure



66. Two circles C_1, C_2 of radii $\frac{1}{3}$ and $\frac{2}{7}$ touch each other externally and they both touch a unit circle C internally. A circle C_3 of radius R is inscribed to touch the circles C_1, C_2 externally and the circle C internally, then 19R is equal to _____

Ans. 2

Sol. $R = \dfrac{\frac{1}{3}\cdot\frac{2}{7}}{1-\frac{1}{3}\cdot\frac{2}{7}} = \frac{2}{19}$

SECTION – D
(Numerical Answer Type)

This section contains **3 questions**. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

67. If the exhaustive set of $x \in (0, 2\pi)$ for which $\forall$ n the inequality $\sin x + \sin 2x + \sin 3x + ... + \sin(nx) \le \frac{\sqrt{3}}{2}$ is valid $\ell_1 \le x \le \ell_2$, $k \in I$, then $\frac{3|\ell_1 - \ell_2|}{2\pi}$ is _____

Ans. **00002.00**

Sol. Clearly, $0 < x < \frac{2\pi}{3}$ not valid, for $\frac{2\pi}{3} \le x \le 2\pi$

$$\sin x = \frac{\cos\left(\frac{x}{2}\right) - \cos\left(\frac{2x+1}{2}\right)x}{2\sin\left(\frac{x}{2}\right)} \le \frac{\cos\left(\frac{x}{2}\right)+1}{2\sin\left(\frac{x}{2}\right)} = \frac{\cot\left(\frac{x}{4}\right)}{2} \le \frac{\sqrt{3}}{2}$$

68. The least numbers A such that for any two squares of combined area 1 and a rectangle, of area A exists such that the two squares can be packed in the rectangle (without interior overlap). You may assume that the sides of the squares are parallel to the sides of the rectangle, is _____

Ans. **00001.20**

Sol. If x and y are the sides of two squares then $x^2 + y^2 = 1$
Let without loss of generality $x \ge y$
So, shorter side of the rectangle should be at least x
Longer side at least and (x + y)
So, $A \equiv x(x + y)$

Put $x = \cos\theta$, $y = \sin\theta$, $\theta \in \left[0, \frac{\pi}{4}\right]$

$A = \frac{1}{2} + \frac{\sqrt{2}}{2}\cos\left(2\theta - \frac{\pi}{4}\right) \le \frac{1+\sqrt{2}}{2}$ equality holds if $\theta = \frac{\pi}{8}$

69. The least possible area of a convex set in plane that intersects both branches of the hyperbola xy = 1 and both branches of the hyperbola xy = –1 is _____

Ans. **00004.00**

Sol. For minimum area x = y = z = w
$A_{min} = 4$

